# BUSINESS REPORT

# TIME SERIES FORECASTING

# (ROSE)

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G4 PGDSBA.O.May 23.A

# Contents

Problem

## Problem:

The data of different types of wine sales in the 20th century is to be analyzed. Both of these data are from the same company but of different wines. As an analyst in the ABC Estate Wines, you are tasked to analyse and forecast Wine Sales in the 20th century.

Data set for the Problem: Rose.csv

### 1. Read the data as an appropriate Time Series Data and plot the Data.

All the necessary packages for analysis of Time Series have been downloaded and data of Rose Wine Sales having 185 rows ranging from 01.01.1980 to 01.07.1995 has been read as Time Series. The data has been checked for missing values and it has been found that there are Two missing values.

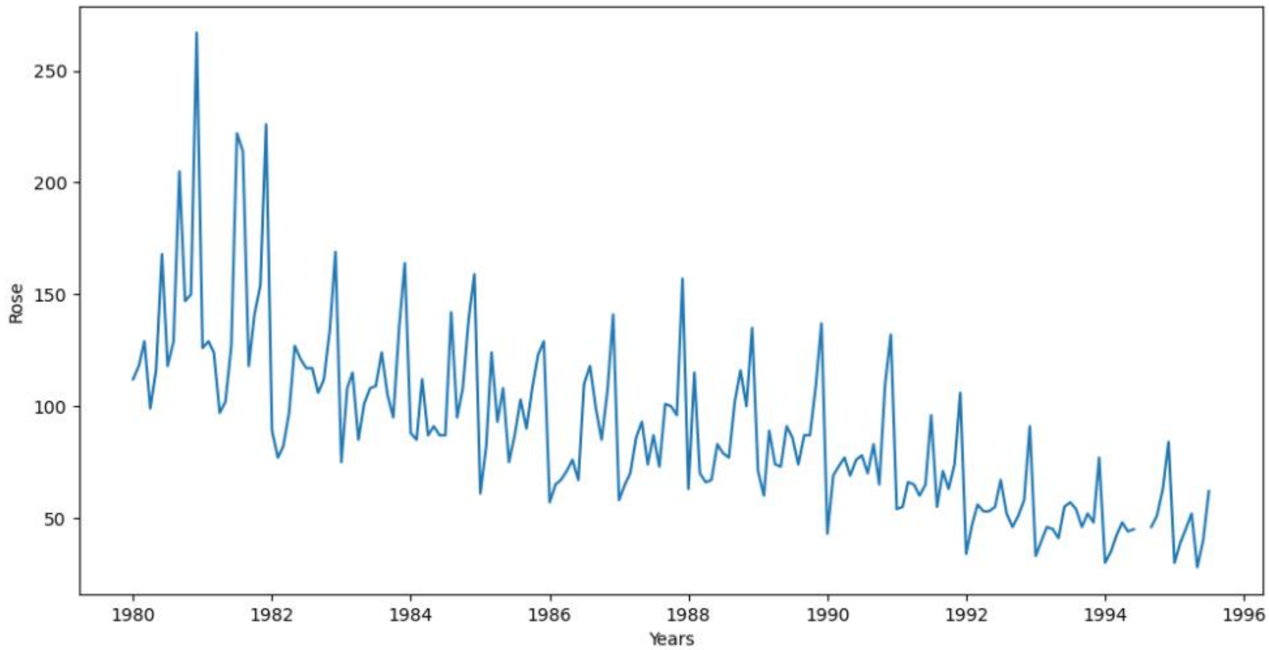
Info of Data:

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 187 entries, 1980-01-01 to 1995-07-01
Data columns (total 1 columns):
 #   Column  Non-Null Count  Dtype  
---  ------  --------------  -----  
 0   Rose    185 non-null    float64
dtypes: float64(1)
memory usage: 2.9 KB
```

Description of Data:

| | Rose |
|---|---|
| count | 185.000000 |
| mean | 90.394595 |
| std | 39.175344 |
| min | 28.000000 |
| 25% | 63.000000 |
| 50% | 86.000000 |
| 75% | 112.000000 |
| max | 267.000000 |

Plotting the data as Time Series:




2. **Perform appropriate Exploratory Data Analysis to understand the data and also perform Decomposition.**

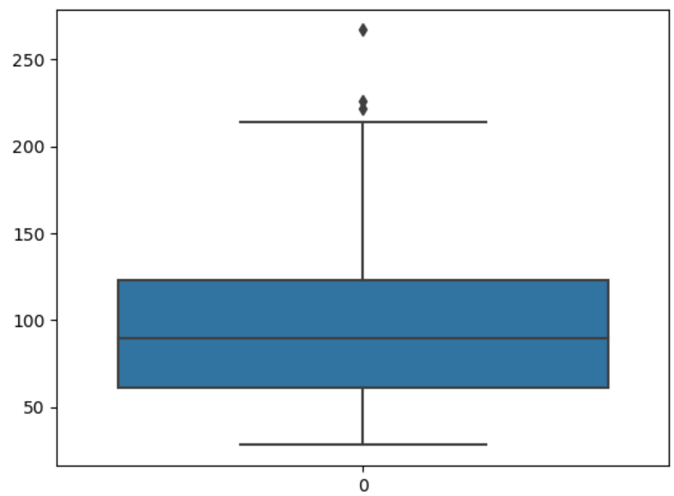
**Missing Values:**

It has been observed that there are two missing values in the data. These two values have been filled with Forward Fill (ffill) method.

**Outliers and Duplicates:**

The data was checked for outliers by plotting boxplot of Rose data. It has been seen that there are some outliers in the data. It has been decided not to drop or impute outliers since they provide important information regarding Wine Sales in different seasons.

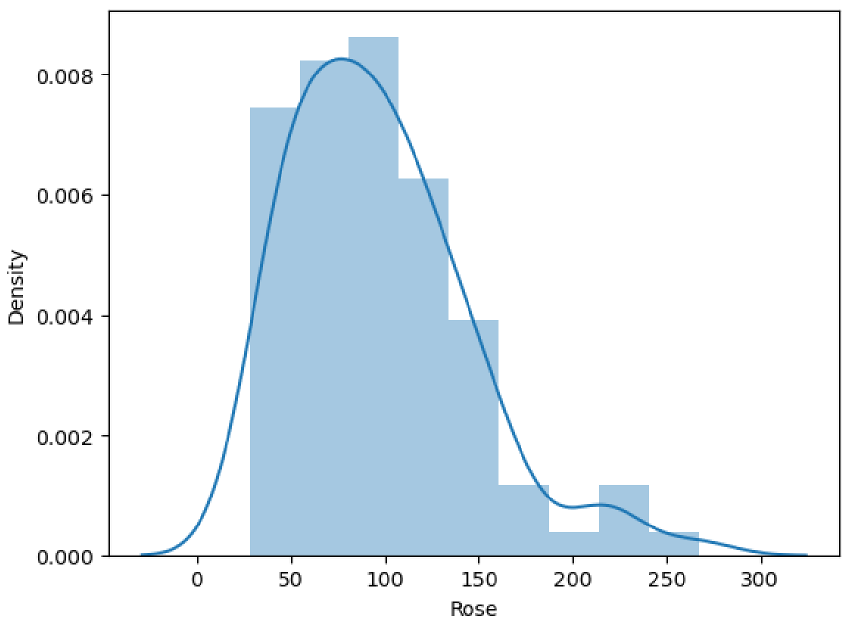
The data has been checked for duplicates and duplicate, if any in the data have been dropped.




**Univariate Analysis:**

The Histogram of Rose Sales have been plotted and it is seen that the data is generally a Normal Distribution with a tail on the right side indicating some outliers.

The Histogram is as under:

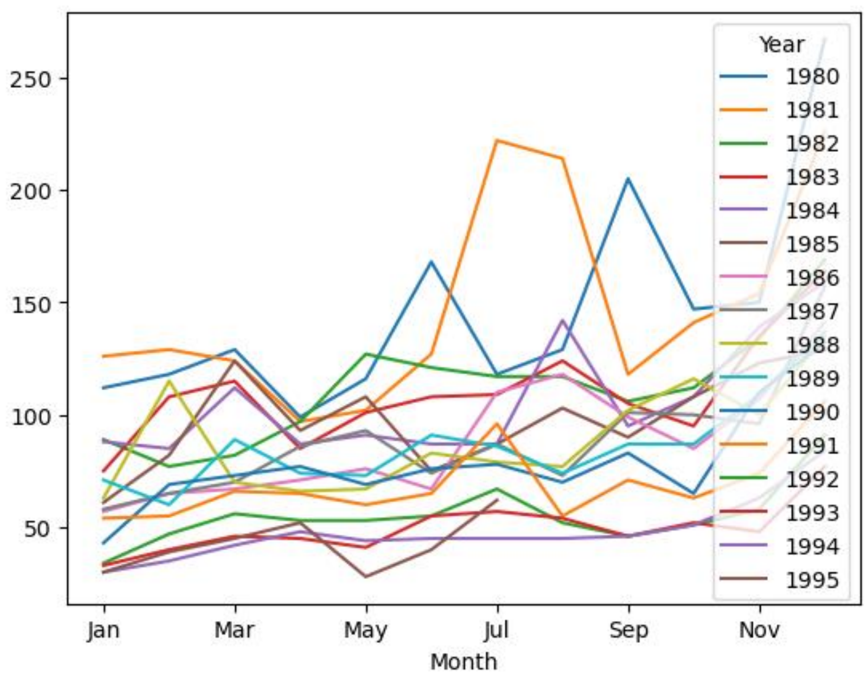


**<u>Bivariate Analysis</u>:**

Pivot and Line Charts of Monthly and Yearly Sales of Rose Wine have been prepared. Further, boxplot of Yearly Sales has also been prepared and results are as s under**:**

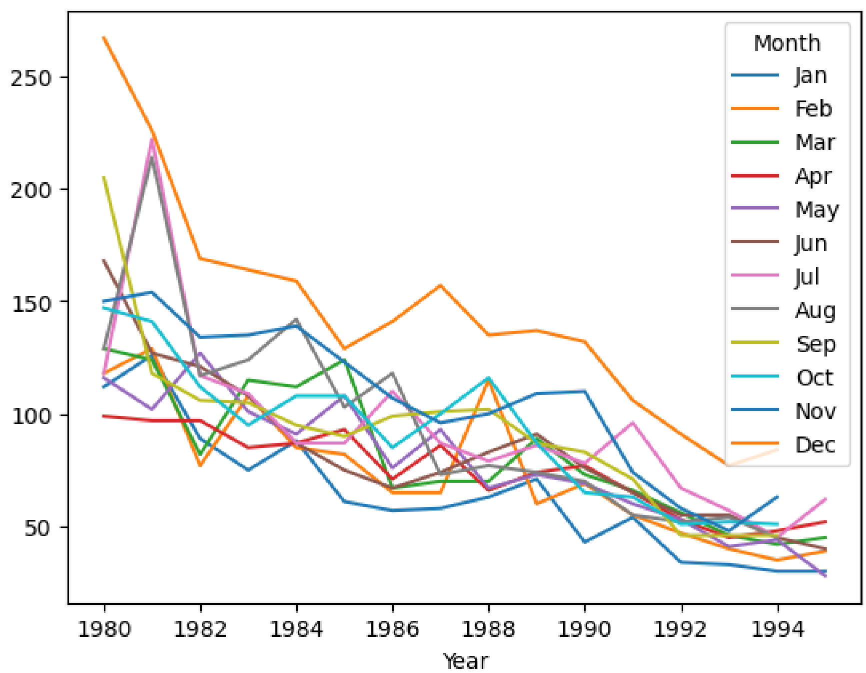
i) Monthly Sales Data:

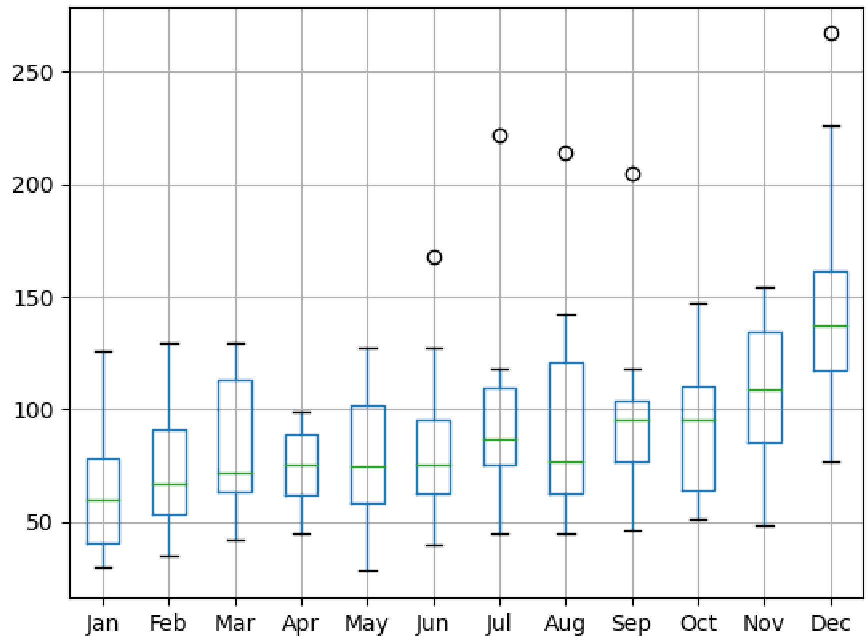
| Year<br>Month | 1980 | 1981 | 1982 | 1983 | 1984 | 1985 | 1986 | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Jan** | 112.0 | 126.0 | 89.0 | 75.0 | 88.0 | 61.0 | 57.0 | 58.0 | 63.0 | 71.0 | 43.0 | 54.0 | 34.0 | 33.0 | 30.0 | 30.0 |
| **Feb** | 118.0 | 129.0 | 77.0 | 108.0 | 85.0 | 82.0 | 65.0 | 65.0 | 115.0 | 60.0 | 69.0 | 55.0 | 47.0 | 40.0 | 35.0 | 39.0 |
| **Mar** | 129.0 | 124.0 | 82.0 | 115.0 | 112.0 | 124.0 | 67.0 | 70.0 | 70.0 | 89.0 | 73.0 | 66.0 | 56.0 | 46.0 | 42.0 | 45.0 |
| **Apr** | 99.0 | 97.0 | 97.0 | 85.0 | 87.0 | 93.0 | 71.0 | 86.0 | 66.0 | 74.0 | 77.0 | 65.0 | 53.0 | 45.0 | 48.0 | 52.0 |
| **May** | 116.0 | 102.0 | 127.0 | 101.0 | 91.0 | 108.0 | 76.0 | 93.0 | 67.0 | 73.0 | 69.0 | 60.0 | 53.0 | 41.0 | 44.0 | 28.0 |
| **Jun** | 168.0 | 127.0 | 121.0 | 108.0 | 87.0 | 75.0 | 67.0 | 74.0 | 83.0 | 91.0 | 76.0 | 65.0 | 55.0 | 55.0 | 45.0 | 40.0 |
| **Jul** | 118.0 | 222.0 | 117.0 | 109.0 | 87.0 | 87.0 | 110.0 | 87.0 | 79.0 | 86.0 | 78.0 | 96.0 | 67.0 | 57.0 | 45.0 | 62.0 |
| **Aug** | 129.0 | 214.0 | 117.0 | 124.0 | 142.0 | 103.0 | 118.0 | 73.0 | 77.0 | 74.0 | 70.0 | 55.0 | 52.0 | 54.0 | 45.0 | NaN |
| **Sep** | 205.0 | 118.0 | 106.0 | 105.0 | 95.0 | 90.0 | 99.0 | 101.0 | 102.0 | 87.0 | 83.0 | 71.0 | 46.0 | 46.0 | 46.0 | NaN |
| **Oct** | 147.0 | 141.0 | 112.0 | 95.0 | 108.0 | 108.0 | 85.0 | 100.0 | 116.0 | 87.0 | 65.0 | 63.0 | 51.0 | 52.0 | 51.0 | NaN |
| **Nov** | 150.0 | 154.0 | 134.0 | 135.0 | 139.0 | 123.0 | 107.0 | 96.0 | 100.0 | 109.0 | 110.0 | 74.0 | 58.0 | 48.0 | 63.0 | NaN |
| **Dec** | 267.0 | 226.0 | 169.0 | 164.0 | 159.0 | 129.0 | 141.0 | 157.0 | 135.0 | 137.0 | 132.0 | 106.0 | 91.0 | 77.0 | 84.0 | NaN |



i) Yearly Sales Data:

| Month | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Year** | | | | | | | | | | | | |
| **1980** | 112.0 | 118.0 | 129.0 | 99.0 | 116.0 | 168.0 | 118.0 | 129.0 | 205.0 | 147.0 | 150.0 | 267.0 |
| **1981** | 126.0 | 129.0 | 124.0 | 97.0 | 102.0 | 127.0 | 222.0 | 214.0 | 118.0 | 141.0 | 154.0 | 226.0 |
| **1982** | 89.0 | 77.0 | 82.0 | 97.0 | 127.0 | 121.0 | 117.0 | 117.0 | 106.0 | 112.0 | 134.0 | 169.0 |
| **1983** | 75.0 | 108.0 | 115.0 | 85.0 | 101.0 | 108.0 | 109.0 | 124.0 | 105.0 | 95.0 | 135.0 | 164.0 |
| **1984** | 88.0 | 85.0 | 112.0 | 87.0 | 91.0 | 87.0 | 87.0 | 142.0 | 95.0 | 108.0 | 139.0 | 159.0 |
| **1985** | 61.0 | 82.0 | 124.0 | 93.0 | 108.0 | 75.0 | 87.0 | 103.0 | 90.0 | 108.0 | 123.0 | 129.0 |
| **1986** | 57.0 | 65.0 | 67.0 | 71.0 | 76.0 | 67.0 | 110.0 | 118.0 | 99.0 | 85.0 | 107.0 | 141.0 |
| **1987** | 58.0 | 65.0 | 70.0 | 86.0 | 93.0 | 74.0 | 87.0 | 73.0 | 101.0 | 100.0 | 96.0 | 157.0 |
| **1988** | 63.0 | 115.0 | 70.0 | 66.0 | 67.0 | 83.0 | 79.0 | 77.0 | 102.0 | 116.0 | 100.0 | 135.0 |
| **1989** | 71.0 | 60.0 | 89.0 | 74.0 | 73.0 | 91.0 | 86.0 | 74.0 | 87.0 | 87.0 | 109.0 | 137.0 |
| **1990** | 43.0 | 69.0 | 73.0 | 77.0 | 69.0 | 76.0 | 78.0 | 70.0 | 83.0 | 65.0 | 110.0 | 132.0 |
| **1991** | 54.0 | 55.0 | 66.0 | 65.0 | 60.0 | 65.0 | 96.0 | 55.0 | 71.0 | 63.0 | 74.0 | 106.0 |
| **1992** | 34.0 | 47.0 | 56.0 | 53.0 | 53.0 | 55.0 | 67.0 | 52.0 | 46.0 | 51.0 | 58.0 | 91.0 |
| **1993** | 33.0 | 40.0 | 46.0 | 45.0 | 41.0 | 55.0 | 57.0 | 54.0 | 46.0 | 52.0 | 48.0 | 77.0 |
| **1994** | 30.0 | 35.0 | 42.0 | 48.0 | 44.0 | 45.0 | 45.0 | 45.0 | 46.0 | 51.0 | 63.0 | 84.0 |
| **1995** | 30.0 | 39.0 | 45.0 | 52.0 | 28.0 | 40.0 | 62.0 | NaN | NaN | NaN | NaN | NaN |

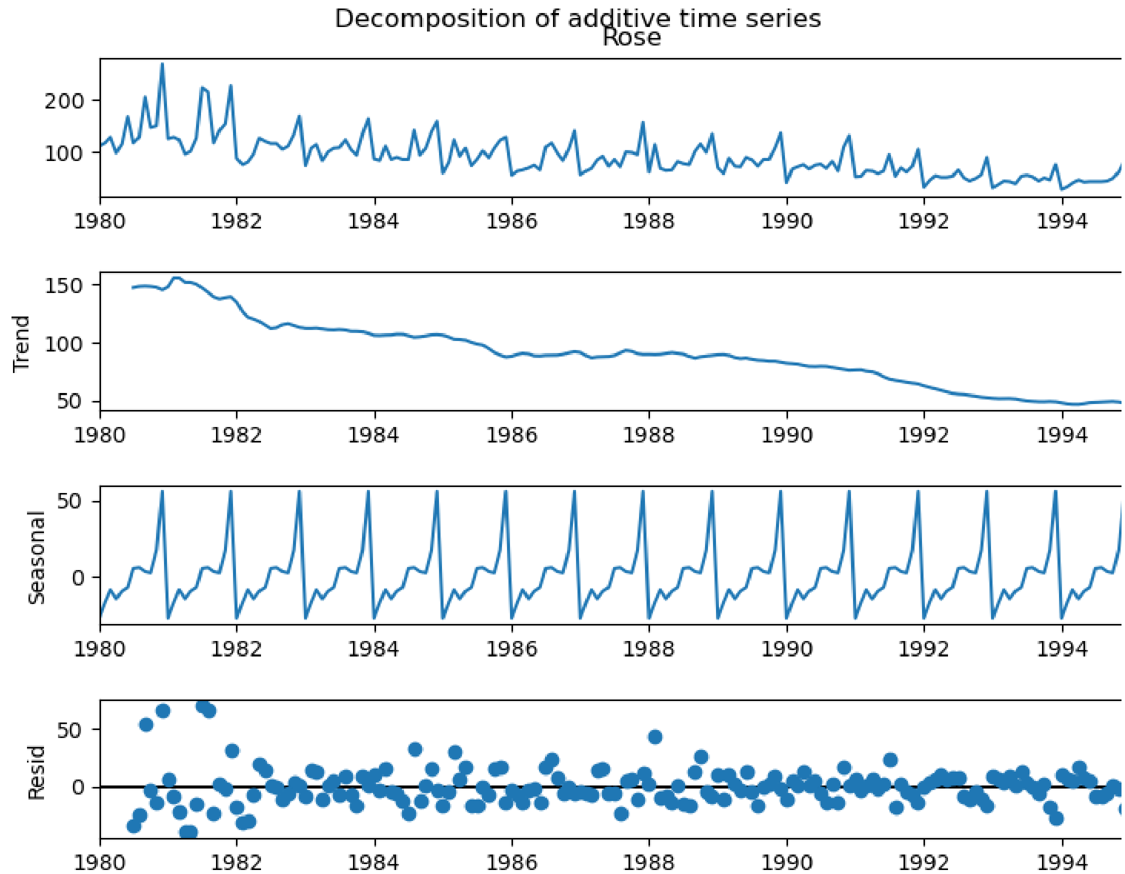


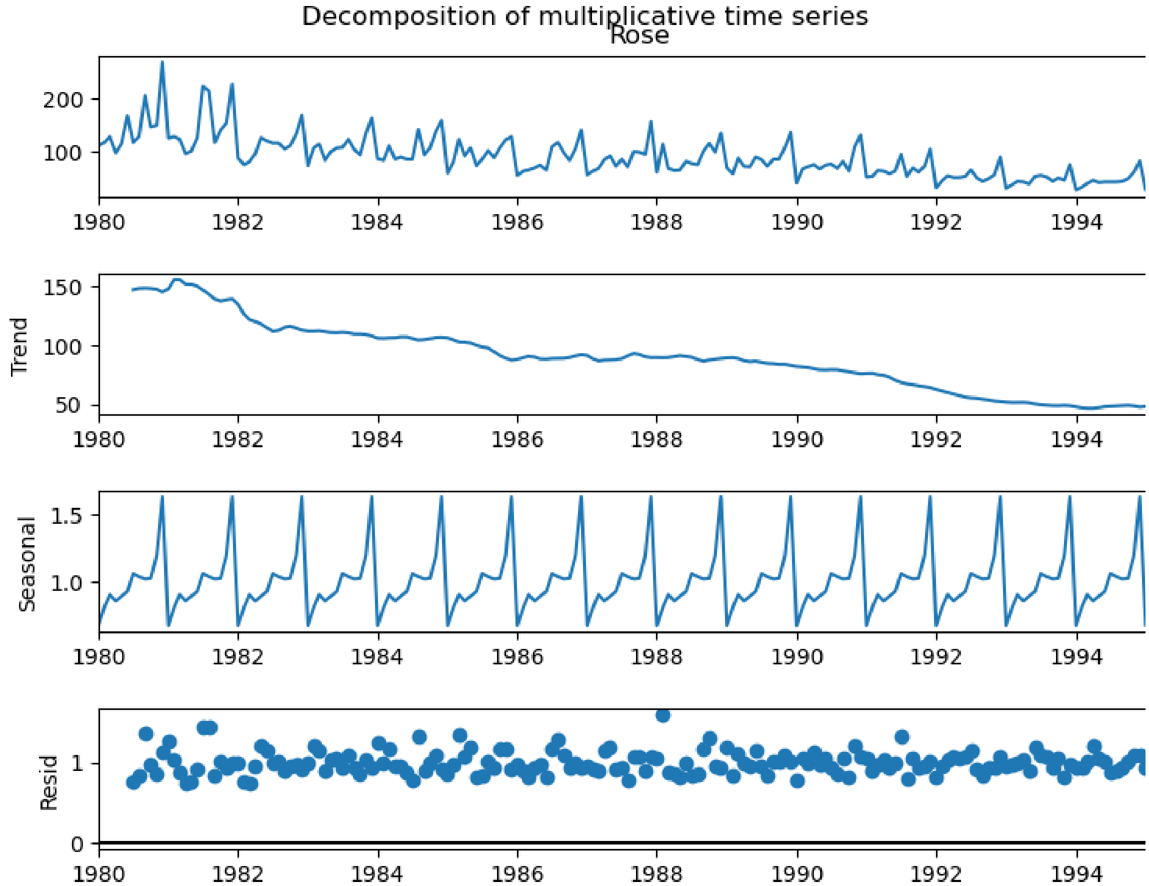



It has been observed that maximum sales are in the months of November and December and minimum sales are in the months January and February. There is prima facie no trend of Year on Year increase in sales.

## **Decomposition Plots:**

The data has been Decomposed for both additive and multiplicative Time Series and the plots are as under:






It may be observed from above that the data has both Trend and Seasonality.

### 3. Split the Data into training and test. The test data should start in 1991.

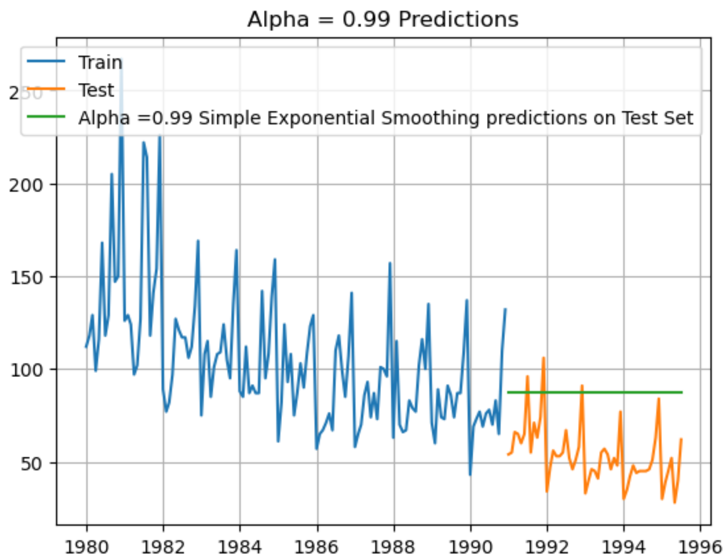
The data has been Split into Training and Test set with Test data starting from 1991. The details are available in Jupyter file appended.

4. **Build all the exponential smoothing models on the training data and evaluate the model using RMSE on the test data. Other additional models such as regression, naïve forecast models, simple average models, moving average models should also be built on the training data and check the performance on the test data using RMSE.**

**Exponential Smoothing Models:**

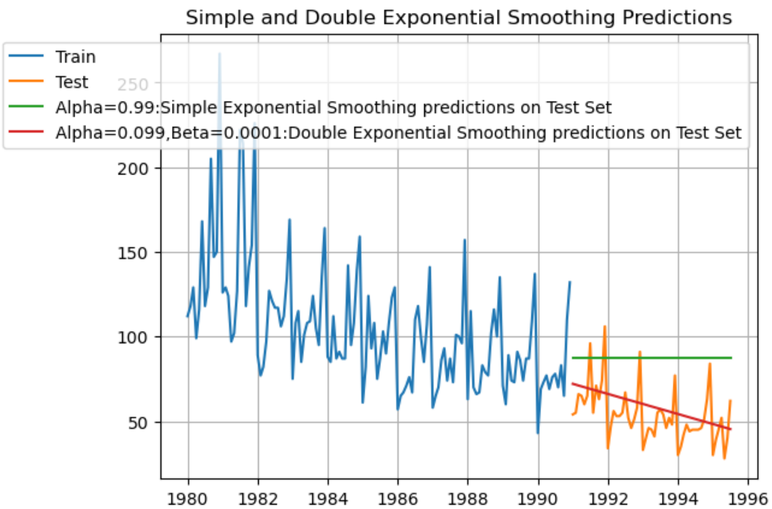
The following Exponential Smoothing Models have been built:

i) Simple Exponential Smoothing: SES was applied on the data. The predictions on test data and RMSE are as under:



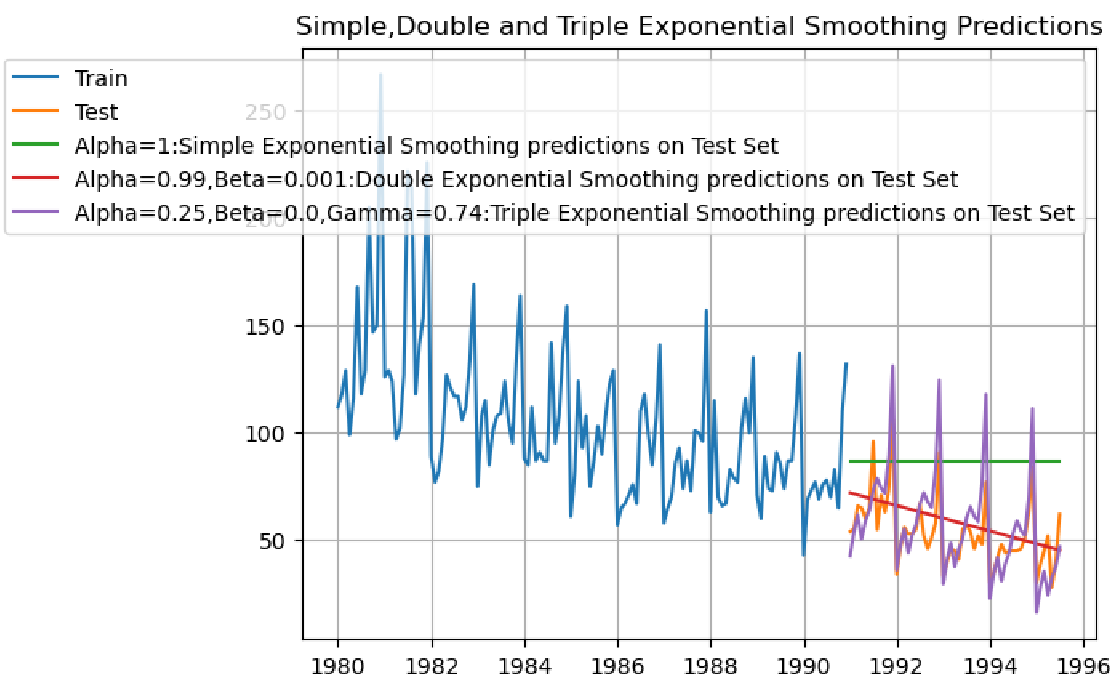

| | Test RMSE |
|---|---|
| Alpha=0.99,SES | 36.81689 |

ii) Double Exponential Smoothing: Double Exponential Smoothing (Holt's Method) was applied on the data. The predictions on test data and RMSE are as under:



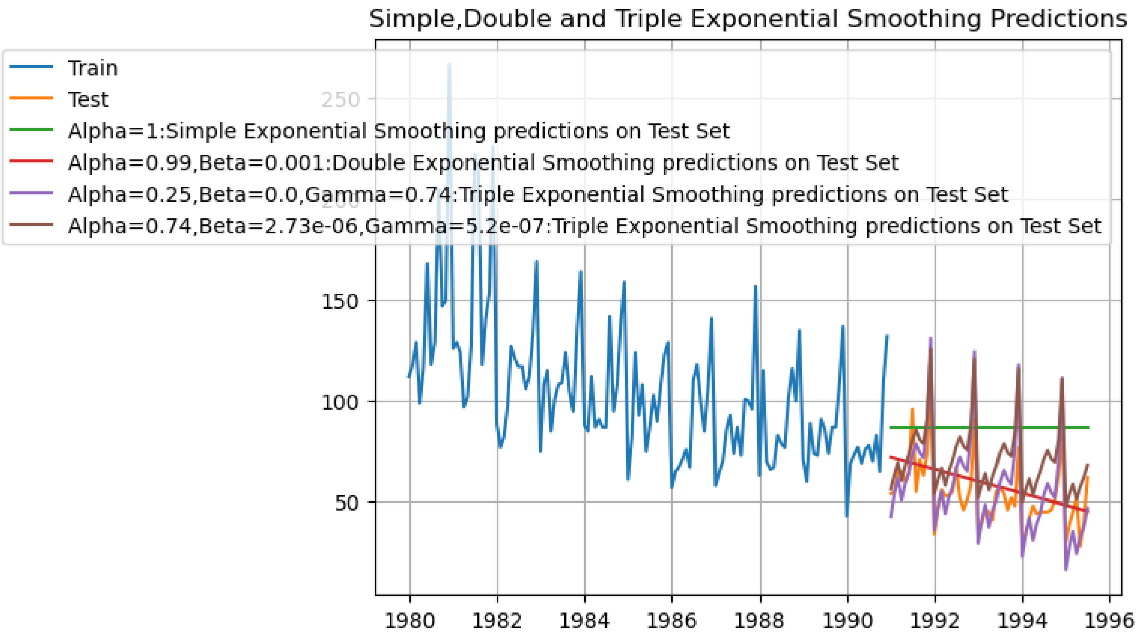

| | Test RMSE |
|---|---|
| **Alpha=0.99,SES** | 36.816890 |
| **Alpha=1,Beta=0.0189:DES** | 15.276106 |

iii) Triple Exponential Smoothing (with additive errors): Triple Exponential Smoothing (Holt Winter's Method) was applied on the data . The predictions on test data and RMSE are as under:



| | Test RMSE |
|---|---|
| Alpha=0.99,SES | 36.816890 |
| Alpha=1,Beta=0.0189:DES | 15.276106 |
| Alpha=0.25,Beta=0.0,Gamma=0.74:TES | 14.280860 |

iv) <u>Triple Exponential Smoothing (with multiplicative errors)</u>: Triple Exponential Smoothing (Holt Winter's Method) was applied on the data The predictions on test data and RMSE are as under:



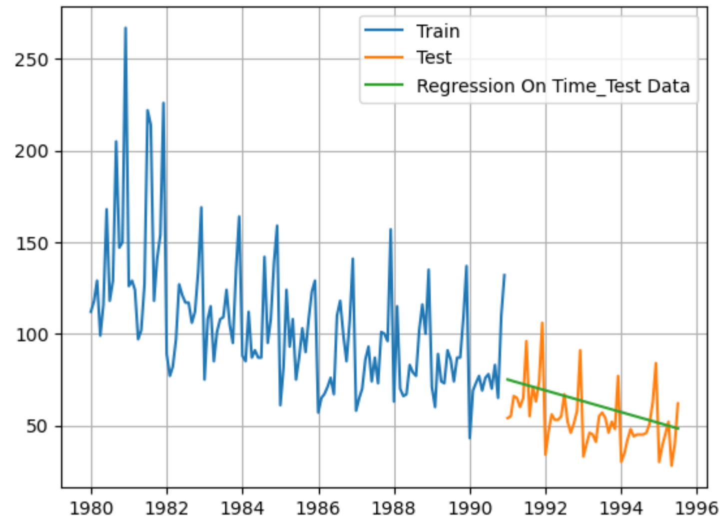

97]:

| | Test RMSE |
|---|---|
| Alpha=0.99,SES | 36.816890 |
| Alpha=1,Beta=0.0189:DES | 15.276106 |
| Alpha=0.25,Beta=0.0,Gamma=0.74:TES | 14.280860 |
| Alpha=0.74,Beta=2.73e-06,Gamma=5.2e-07,Gamma=0:TES | 20.216957 |
| Alpha=0.74,Beta=2.73e-06,Gamma=5.2e-07,Gamma=0:TES | 20.216957 |

**<u>Other Additional Models</u>:**

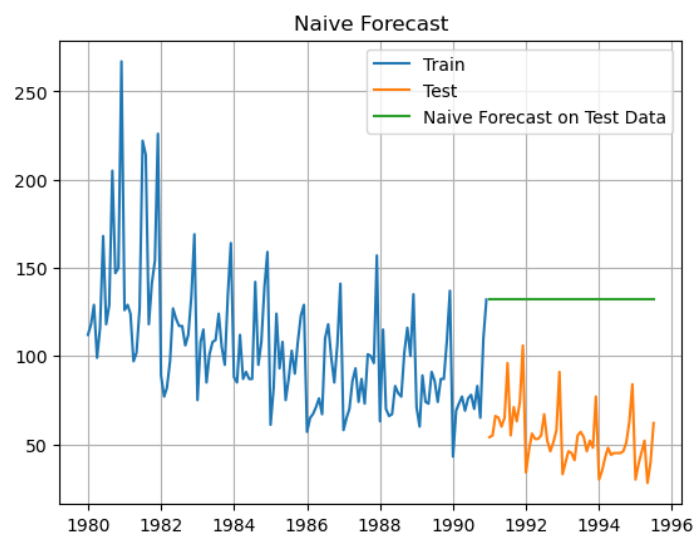
The following Other Additional Models have been built:

i) <u>Linear Regression Model:</u> Linear Regression was applied on the data. The predictions on test data and RMSE are as under:



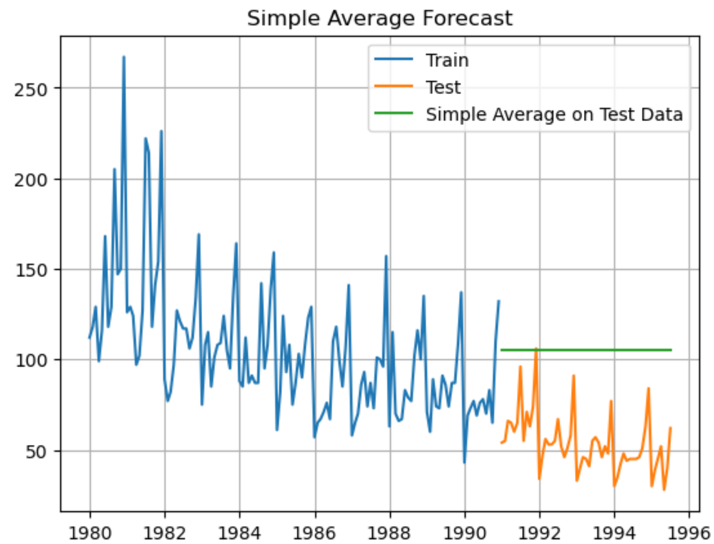

| | Test RMSE |
|---|---|
| RegressionOnTime | 16.464804 |

ii) <u>Naïve Forecast Model:</u> Naïve Forecast Model was applied on the data The predictions on test data and RMSE are as under:



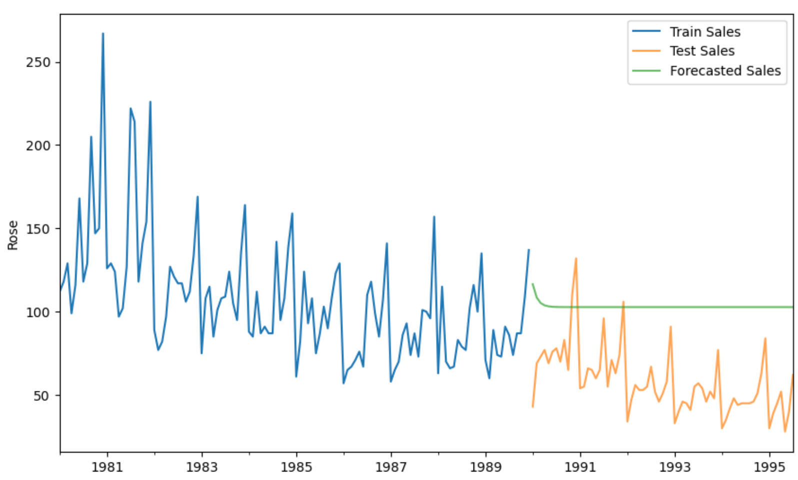
| | Test RMSE |
|---|---|
| RegressionOnTime | 16.464804 |
| NaiveModel | 79.738550 |

iii) Simple Average Model: Simple Average Model was applied on the data The predictions on test data and RMSE are as under:




| | Test RMSE |
|---|---|
| RegressionOnTime | 16.464804 |
| NaiveModel | 79.738550 |
| SimpleAverageModel | 53.480857 |

iv) Auto Regressive Model: AR Model was built on the data. The predictions on test data and RMSE are as under:

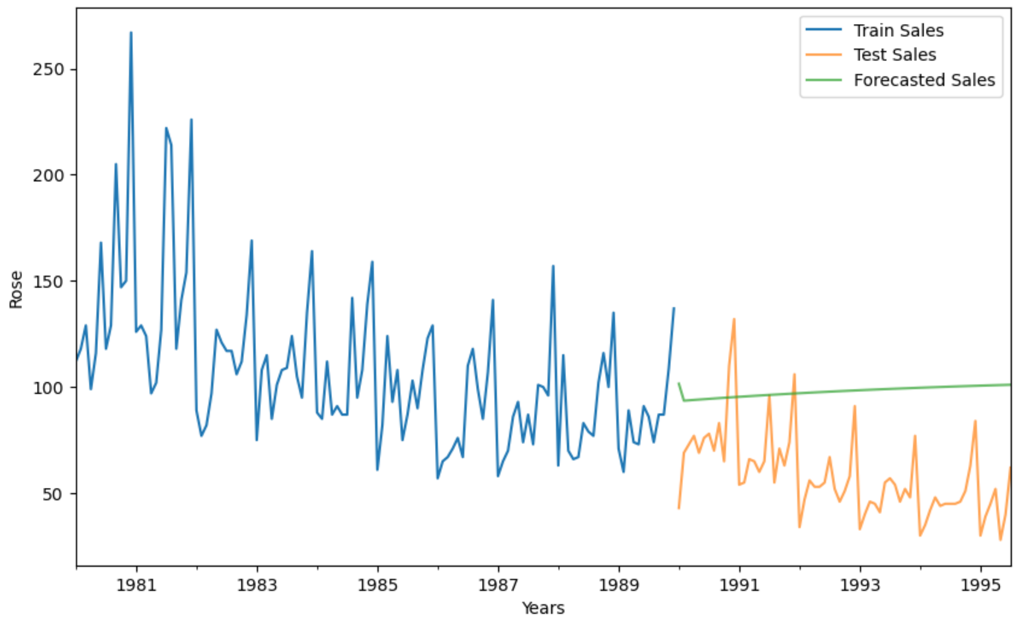


:

| | RMSE |
|---|---|
| Best AR Model : ARIMA(2,0,0) | 48.891405 |

We can observe a flat output because AR model expresses the current value as a linear combination of past values.

v) Auto Regressive Moving Average Model: ARMA Model was built on the data. The predictions on test data and RMSE are as under:




]:

| | RMSE |
|---|---|
| Best AR Model : ARIMA(2,0,0) | 48.891405 |
| Best ARMA Model : ARIMA(3,0,3) | 44.794385 |

From the output, we can observe that ARMA Model combines both Auto Regressive and Moving Average terms.

**5. Check for the stationarity of the data on which the model is being built on using appropriate statistical tests and also mention the hypothesis for the statistical test. If the data is found to be non-stationary, take appropriate steps to make it stationary. Check the new data for stationarity and comment.**
**Note: Stationarity should be checked at alpha = 0.05.**

The stationarity of data has been checked using Dicky Fuller Test:

Null Hypothesis is that the Series is non-stationary and

Alternate Hypothesis is that the series is stationary.

The Test has been done at Alpha = 0.05. The results are as under:

```
# Dickey-Fuller Test - Dicky Fuller Test on the timeseries is run to check for stationarity of data.

# Null Hypothesis H0: Time Series is non-stationary.
# Alternate Hypothesis Ha: Time Series is stationary.
#So Ideally if p-value < 0.05 then null hypothesis: TS is non-stationary is rejected else
# the TS is non-stationary is failed to be rejected
```

```
DF test statistic is -1.157
DF test p-value is 0.6917
```

```
# Since p value is greater than 0.05, therefore. we accept the Null Hypothesis and Time Series is non Stationary.
```

In this case, p value is 0.6917, which is greater than 0.05, therefore, we fail to reject the Null Hypothesis and conclude that the Series is non-stationary.

We make the series stationary using differencing. After differencing the Time Series, Dicky Fuller Test for Stationarity is run. The results of Dicky Fuller Test are as under:

```
DF test statistic is -8.247
DF test p-value is 0.0000
```

Since, p value is less than alpha (0.05), we reject the Null Hypothesis and the Time Series is now stationary.

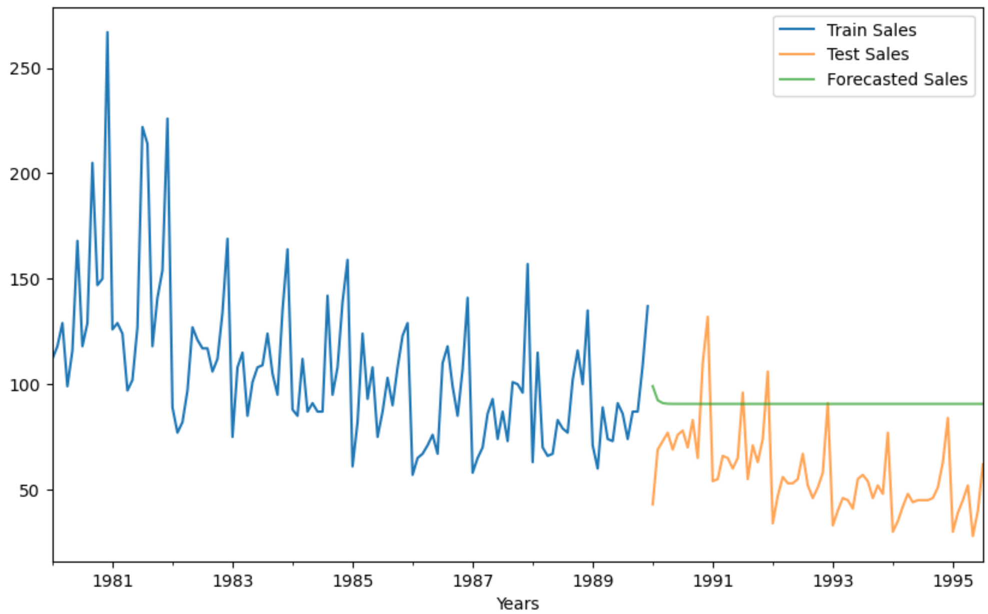
6. **Build an automated version of the ARIMA/SARIMA model in which the parameters are selected using the lowest Akaike Information Criteria (AIC) on the training data and evaluate this model on the test data using RMSE.**

Automated versions of ARIMA and SARIMA Models wherein parameters were selected using the lowest Akaike Information Criteria (AIC) built on the training data and the models were evaluated on the test data. The results are as under:

**ARIMA Model**: Best Results using lowest AIC of ARIMA Model, Predictions and Root Mean Square Error are as under:

| | param | AIC |
|---|---|---|
| 3 | (1, 1, 1) | -172.217295 |
| 4 | (1, 1, 2) | -172.080853 |
| 1 | (1, 0, 2) | -171.990139 |
| 6 | (2, 0, 1) | -171.251383 |
| 7 | (2, 0, 2) | -171.092882 |

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                   Rose   No. Observations:                  120
Model:                 ARIMA(1, 1, 1)   Log Likelihood                  89.109
Date:                Sat, 27 Jan 2024   AIC                           -172.217
Time:                        16:23:39   BIC                           -163.880
Sample:                    01-01-1980   HQIC                          -168.832
                         - 12-01-1989
Covariance Type:                  opg
==============================================================================
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.2149      0.087      2.468      0.014       0.044       0.386
ma.L1         -0.9198      0.055    -16.851      0.000      -1.027      -0.813
sigma2         0.0129      0.002      7.428      0.000       0.010       0.016
==============================================================================
```



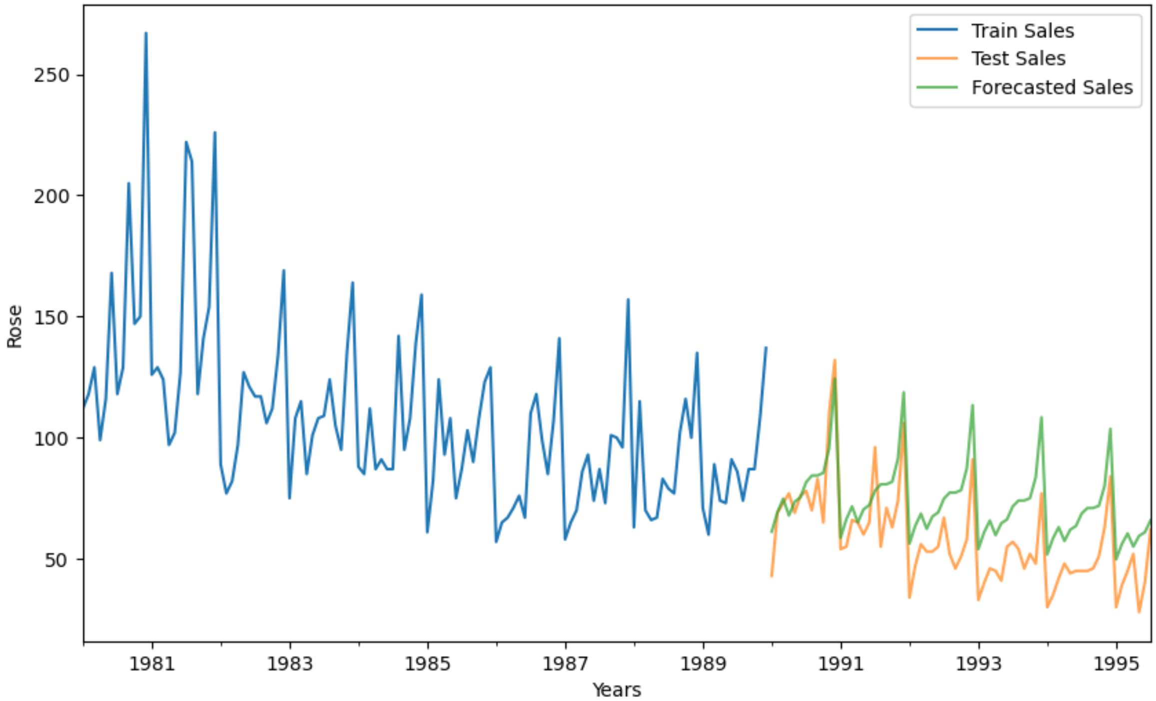

| | RMSE |
|---|---|
| Best AR Model : ARIMA(2,0,0) | 48.891405 |
| Best ARMA Model : ARIMA(3,0,3) | 44.794385 |
| Best ARIMA Model : ARIMA(3,0,3) | 37.961345 |

**<u>SARIMA Model</u>**: Best Results using lowest AIC of SARIMA Model, Predictions and Root Mean Square Error are as under:

| | param | seasonal | AIC |
|---|---|---|---|
| **0** | (1, 0, 1) | (1, 0, 1, 12) | -224.610695 |
| **108** | (2, 0, 1) | (1, 0, 1, 12) | -222.981478 |
| **126** | (2, 0, 2) | (1, 0, 1, 12) | -218.826363 |
| **162** | (2, 1, 1) | (1, 0, 1, 12) | -218.233124 |
| **18** | (1, 0, 2) | (1, 0, 1, 12) | -218.182615 |

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                               Rose   No. Observations:                  120
Model:             SARIMAX(1, 0, 1)x(1, 0, 1, 12)   Log Likelihood                 122.469
Date:                            Sat, 27 Jan 2024   AIC                           -234.937
Time:                                    17:11:32   BIC                           -221.000
Sample:                                01-01-1980   HQIC                          -229.277
                                     - 12-01-1989
Covariance Type:                              opg
==========================================================================================
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.9994      0.001    777.317      0.000       0.997       1.002
ma.L1         -0.9053      0.050    -18.116      0.000      -1.003      -0.807
ar.S.L12       0.9942      0.013     75.960      0.000       0.969       1.020
ma.S.L12      -0.8718      0.136     -6.409      0.000      -1.138      -0.605
sigma2         0.0059      0.001      6.308      0.000       0.004       0.008
==============================================================================
```

| | Year-Month | Rose | Month | Year | date | Rose_sales_forecasted |
|---|---|---|---|---|---|---|
| **1990-01-01** | 1990-01 | 43.0 | Jan | 1990 | 1990-01-01 | 61.195107 |
| **1990-02-01** | 1990-02 | 69.0 | Feb | 1990 | 1990-02-01 | 69.199679 |
| **1990-03-01** | 1990-03 | 73.0 | Mar | 1990 | 1990-03-01 | 74.780532 |
| **1990-04-01** | 1990-04 | 77.0 | Apr | 1990 | 1990-04-01 | 67.875402 |
| **1990-05-01** | 1990-05 | 69.0 | May | 1990 | 1990-05-01 | 73.509815 |
| **...** | ... | ... | ... | ... | ... | ... |
| **1995-03-01** | 1995-03 | 45.0 | Mar | 1995 | 1995-03-01 | 60.452085 |
| **1995-04-01** | 1995-04 | 52.0 | Apr | 1995 | 1995-04-01 | 55.025600 |
| **1995-05-01** | 1995-05 | 28.0 | May | 1995 | 1995-05-01 | 59.460509 |
| **1995-06-01** | 1995-06 | 40.0 | Jun | 1995 | 1995-06-01 | 60.921075 |
| **1995-07-01** | 1995-07 | 62.0 | Jul | 1995 | 1995-07-01 | 65.830321 |




| | RMSE |
|---|---|
| Best AR Model : ARIMA(2,0,0) | 48.891405 |
| Best ARMA Model : ARIMA(3,0,3) | 44.794385 |
| Best ARIMA Model : ARIMA(3,0,3) | 37.961345 |
| Best SARIMA Model : SARIMAX(1, 0, 1)x(1, 0, 1, 12) | 18.419326 |

**It is observed that Best SARIMA Model (1,0,1)x(1,0,1,12) with RMSE of 18.42 is the Best Model.**

**7. Build a table with all the models built along with their corresponding parameters and the respective RMSE values on the test data.**

**Comparison of Models:** Comparison of all the aforesaid Models is tabulated as under**:**

| **S.No.** | **Model** | **Root Mean Square Error (RMSE) on Test Data** |
|---|---|---|
| **1.** | Simple Exponential Smoothing Model | 36.82 |
| **2.** | Double Exponential Smoothing Model | 15.28 |
| **3.** | Triple Exponential Smoothing (with additive errors) | 14.28 |
| **4.** | Triple Exponential Smoothing (with multiplicative errors) | 20.22 |
| **5.** | Linear Regression | 16.46 |
| **6.** | Naïve Model | 79.74 |
| **7.** | Simple Average Model | 53.48 |
| **8.** | Auto Regressive (AR) Model | 48.89 |
| **9.** | Auto Regressive Moving Average (ARMA) Model | 44.79 |
| **10.** | ARIMA Model | 37.96 |
| **11.** | SARIMA Model | 18.42 |

**It is seen from above that SARIMA Model with Lowest RMSE is the best Model which captures both trend and seasonality in an optimal manner.**

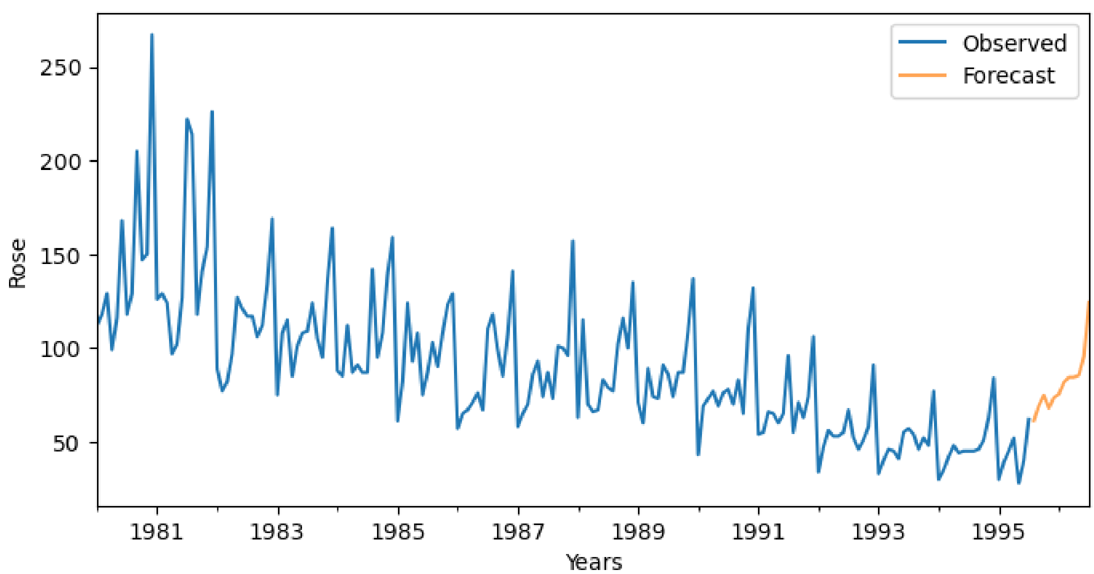
**8. Based on the model-building exercise, build the most optimum model(s) on the complete data and predict 12 months into the future with appropriate confidence intervals/bands.**

Forecasting was done for the 12 months into future using best SARIMA Model with lowest AIC and the results and plot of predictions with 95% Confidence Interval and 99% Confidence Interval is as under:

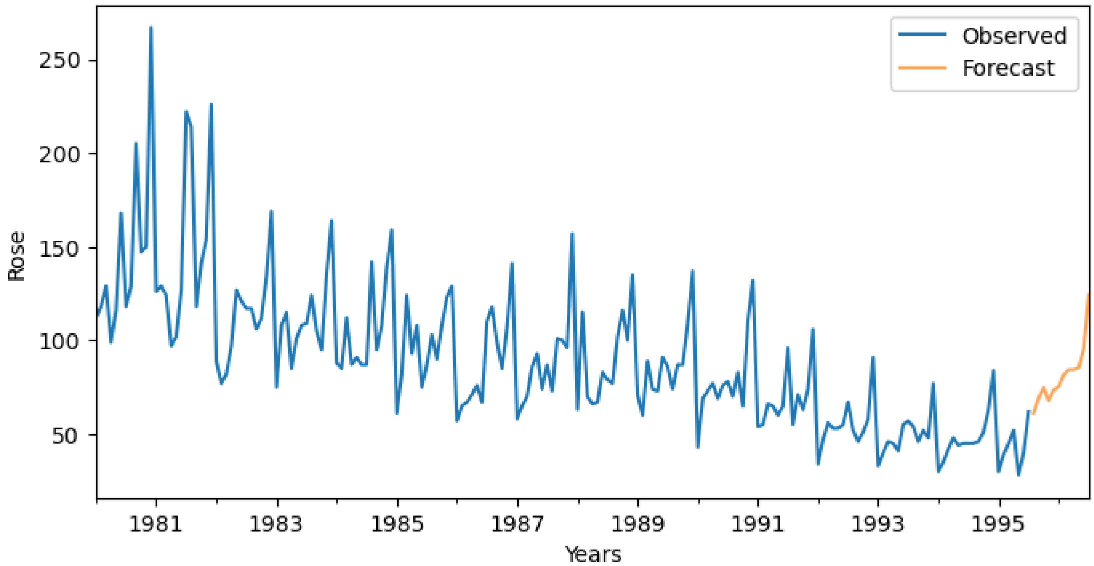
]:

| | forecast | lower_ci_95 | upper_ci_95 | lower_ci_99 | upper_ci_99 |
|---|---|---|---|---|---|
| **1995-08-01** | 61.195107 | 43.114982 | 86.857072 | 43.114982 | 86.857072 |
| **1995-09-01** | 69.199679 | 48.679865 | 98.369122 | 48.679865 | 98.369122 |
| **1995-10-01** | 74.780532 | 52.525671 | 106.464667 | 52.525671 | 106.464667 |
| **1995-11-01** | 67.875402 | 47.603314 | 96.780452 | 47.603314 | 96.780452 |
| **1995-12-01** | 73.509815 | 51.477295 | 104.972355 | 51.477295 | 104.972355 |
| **1996-01-01** | 75.365221 | 52.697600 | 107.783212 | 52.697600 | 107.783212 |
| **1996-02-01** | 81.620330 | 56.986432 | 116.902885 | 56.986432 | 116.902885 |
| **1996-03-01** | 84.363707 | 58.814687 | 121.011186 | 58.814687 | 121.011186 |
| **1996-04-01** | 84.389069 | 58.745830 | 121.225879 | 58.745830 | 121.225879 |
| **1996-05-01** | 85.494556 | 59.428353 | 122.993804 | 59.428353 | 122.993804 |
| **1996-06-01** | 95.579584 | 66.341969 | 137.702530 | 66.341969 | 137.702530 |
| **1996-07-01** | 124.335490 | 86.176705 | 179.390870 | 86.176705 | 179.390870 |

Plot of predictions with 95% Confidence Interval is as under:



Plot of predictions with 99% Confidence Interval is as under:




**9. Comment on the model thus built and report your findings and suggest the measures that the company should be taking for future sales.**

It is seen that SARIMA Model with the lowest AIC of -224.61 and RMSE of 18.42 is the best Model which can predict both trend and seasonality of the data while also taking care of the stationarity factor.

Based on the Model selected, the suggested measures for the Company to be taken for future 12 months sales are as under:

i) It has been seen from predictions that sales of Sparkling Wine start increasing in the Month of October 1995 and continue till July 1996 with peak in the Month of July 1996. It is required for the Company to keep high inventory of raw materials for production from September 1995 onwards.

ii) Adequate planning and deployment of resources is required to meet the demand during peak sales period.

iii) Attractive discounts and promotional activities to be undertaken to boost wine sales in lean months.

iv) Lowest sales are in the month of August 1995. Hence, the routine maintenance of machinery should be planned in August 1995.

v) Export activities may be undertaken during lean period to boost sales